

EECS 182 Cheat Sheet: Muon Optimizer & μP (Maximal Update Parametrization)

1 Muon Optimizer

1.1 Purpose

- Scale-invariant optimization: updates unaffected by rescaling parameters.
- Controls *relative* change of parameter norm.
- Stabilizes deep & wide network training.

1.2 Core Update Rule (Conceptual)

Given parameters θ_t and gradient $g_t = \nabla_{\theta} L(\theta_t)$:

$$u_t = \frac{\|\theta_t\|}{c + \|\theta_t\|} \cdot \frac{g_t}{\|g_t\| + \epsilon}$$

$\theta_{t+1} = \theta_t - \eta u_t$

Key:

- Gradient normalization $\frac{g_t}{\|g_t\|}$ removes scale dependence of gradients.
- Multiplicative factor $\frac{\|\theta_t\|}{c + \|\theta_t\|}$ enforces relative-norm updates.

1.3 Scale Invariance Derivation

Rescale parameters:

$$\theta' = \alpha\theta, \quad g' = \alpha g$$

Then:

$$\|g'\| = \alpha\|g\|, \quad \|\theta'\| = \alpha\|\theta\|$$

Muon direction becomes:

$$u'_t = \frac{\alpha\|\theta\|}{c + \alpha\|\theta\|} \cdot \frac{\alpha g}{\alpha\|g\|} = \frac{\alpha\|\theta\|}{c + \alpha\|\theta\|} \cdot \frac{g}{\|g\|}$$

If $\alpha\|\theta\| \gg c$:

$$u'_t \approx u_t$$

Effective update direction is scale-invariant.

1.4 Relative Norm Change

Approximate:

$$\frac{\|\Delta\theta\|}{\|\theta\|} \approx \eta \frac{1}{\|g_t\|}$$

$\Rightarrow$ Muon behaves like a fixed percentage update.

2 μP : Maximal Update Parametrization

2.1 Purpose

- Ensures training dynamics converge as width $n \rightarrow \infty$.
- Makes small models predictive of large-model behavior.
- Specifies initialization & learning-rate scaling rules for width-invariant updates.

2.2 Network Setup

Hidden width n MLP:

$$h^\ell = \phi(W^\ell h^{\ell-1} + b^\ell), \quad W^\ell \in \mathbb{R}^{n \times n}$$

2.3 μP Initialization (Simplified)

$$W_{ij}^\ell \sim \mathcal{N}(0, \frac{1}{n})$$
$$b^\ell \sim \mathcal{N}(0, 1)$$

2.4 μ P Learning-Rate Scaling

Feature weights:

$$\eta_{W^\ell} \propto \frac{1}{n}$$

Readout weights:

$$\eta_{W^L} = O(1)$$

2.5 Derivation: Why $\eta \propto 1/n$?

Toy model:

$$f(x) = \frac{1}{\sqrt{n}} \sum_{i=1}^n a_i \phi(w_i^\top x)$$

Gradient:

$$\frac{\partial f}{\partial w_i} = \frac{1}{\sqrt{n}} a_i \phi'(w_i^\top x) x$$

Magnitude: $O(\frac{1}{\sqrt{n}})$.

Total functional update:

$$\Delta f(x) \sim \sum_{i=1}^n \frac{1}{\sqrt{n}} \eta_w \sim \eta_w \sqrt{n}$$

Width-invariant update requires:

$$\eta_w \sqrt{n} = O(1) \quad \Rightarrow \quad \eta_w \propto \frac{1}{\sqrt{n}}$$

(Refined in μ P to $\eta \propto \frac{1}{n}$ for feature weights.)

2.6 What μ P Guarantees

With correct init + LR scaling:

$$f_n(x; \theta_t) \rightarrow f_\infty(x; \theta_t), \quad \Delta f_n(x) \rightarrow \Delta f_\infty(x)$$

Training curves align across widths.

3 Muon vs μ P: Summary Table

Concept	Controls	Mathematical Idea
μ P	Width scaling	Keep $\Delta f(x)$ constant across n
Muon	Parameter scaling	Normalize by $\ \theta\ $ and $\ g\ $

4 Exam Checklist

Be able to:

- Prove Muon scale-invariance under $\theta \rightarrow \alpha \theta$.
- Derive LR scaling for n -width MLPs.
- Compare SP, NTK, and μ P parametrizations.
- Explain why Adam is not width-invariant.
- Compute $\Delta f(x)$ for a toy network under a given update rule.